Chenyang Xu

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Education

Courant, New York University **Jan. 2024 – Dec. 2025**

Master of Science, Computer Science

- Computer Vision, Geometric Processing, Computer Graphics(GPA: 3.9)

St John's College, University of Cambridge **Oct. 2017 – Jun. 2021**

Combined Bachelor and Master Program in Engineering, Major in Computational and Information Engineering

- General Engineering in the 1st Year and 2nd Year: mechanical engineering, electrical and electronic engineering, materials and information engineering.
- Information Engineering in the 3rd and 4th Year: control, information theory, reinforcement learning.

Research

Night at the Museum: A Scalable Framework for Text-Driven Mesh Motion Generation (In Submission) **Sep. 2025 - Now**

Supervisor: Chen Feng, AI4CE, NYU

- Designed a skeleton-free framework for text-driven mesh motion generation as the first author.
- Proposed a novel semantic-aware rigging representation and a diffusion-based generation method built upon it. The framework, espacially the rigging, requires no manual manipulation, making it naturally scalable.
- Curated a large-scale animal mesh motion dataset to demonstrate scalability and support further research.

A FEM Solver Benchmark **Feb. 2025 - Now**

Supervisor: Denis Zorin, Teseo Schneider, Daniele Panozzo, Geometric Computing Lab, NYU

- Conducting systematic benchmarks of finite element method (FEM) solvers, evaluating and analyzing the performance of both iterative and direct linear solvers, as the first author.

Crag: Can 3D Generative Models Help 3D Assembly? (Under Review) **Apr. 2025 - Sep. 2025**

Supervisor: Chen Feng, AI4CE, NYU

- Designed a physically based simulation pipeline to model the fossilization process of bones to curate a new dataset, implemented FractureBEM for modeling brittle fracture behavior.

GARF: Learning Generalizable 3D Reassemble for Real-World Fractures (ICCV 2025) **Jan. 2025 - Apr. 2025**

Supervisor: Chen Feng, AI4CE, NYU

- Designed a generalizable flow matching-based 3D reassembly method trained on 1.9 Million fractures, enabling precise real-world fragment pose alignment.

Linear Gaussian Model hyper-parameters Optimization in Successor Uncertainties **Aug. 2020 - Jun. 2021**

Supervisor: José Miguel Hernández Lobato, David Janz, Computational and Biological Learning Lab, University of Cambridge

- Applied the Bayesian linear model to improve the Successor Uncertainties (SU), a state-of-the-art method for efficient exploration in model-free reinforcement learning.

Improvement of Audio Modem **Apr. 2020**

Supervisor: Jossy Sayir, University of Cambridge

- Explored ways to improve the OFDM based audio modem, solved linear phase addition issues due to OFDM block truncation and evaluated it mathematically.

Experience

Wayve, Autonomous Driving, London, UK **Jul. 2019 - Aug. 2019**

Software Engineering Summer Intern

- Built the fleet real-time monitoring system which can supervise the status of autonomous driving vehicles e.g. GPS information. The system played a vital role on data analysis of the fleet.

Deckzero, Software Engineering, London **Jun. 2018 - Sept. 2018**

Software Engineering Summer Intern

- Built the prototype of an education web tool which can transform materials in the form for self-learning, e.g. jupyter notebook, to presentable slices.

Activities

Gamecores, Booom, China **Aug. 2021**

Game Jam, team leader and main programmer

- Made an act-rpg game using Unity, *Twelve Terrestrial Branches*. Two players need to complete their own tasks in their individual world, however, these two worlds seem to have the delicate relationship that reveals a mystery story. (https://www.gcores.com/games/67517)

Cambridge University Chinese Debating Society **Apr. 2018 – Apr. 2019**

Publicity Officer

- Represented university to attend the International Chinese Debating Competition (xin guo bian in Chinese) in 2019.
- Helped hold the biggest inter-university Chinese debating conference around UK in 2019.

Additional Information

Computing Python, C++, C#, HTML, CSS, JavaScript, Linux, Matlab, Blender, Unity

Languages English, Chinese

Academic Challenge Awards

- British Physics Olympiad (BPhO) Final Round Silver 2017 (only the first 50 participants awarded to this round)
- Cambridge Chemistry Challenge (C3L6) Final Round Gold 2016